P3:AI-Driven Land Use Land Cover Classification

Project Report Submitted in fulfilment of the Requirements for the Award of the Internship of Winter Training Program on Space Science Technology

Subject Name:Remote sensing and GIS

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DECLARATION

I hereby declare that the project titled *"AI-Driven Land Use and Land Cover Classification Using Supervised Machine Learning"* is an original work carried out by me during the Winter Training Program conducted by India Space Academy. The work presented in this report has been completed under the academic guidelines provided and has not been submitted elsewhere for any academic certification.

Prajin S

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to India Space Academy for organizing the Winter Training Program on Remote Sensing and GIS with Artificial Intelligence. The structured curriculum and hands-on exposure to tools such as Google Earth Engine and QGIS have significantly enhanced my understanding of satellite data processing and spatial analysis.

I extend my heartfelt thanks to the Course Coordinator for providing valuable guidance and technical insights throughout the duration of the program. The continuous support and detailed explanations during the live sessions helped me build strong foundational and practical knowledge in geospatial analytics.

I would also like to acknowledge the open-source communities of Google Earth Engine and QGIS for providing powerful and accessible tools that enable advanced satellite data analysis. Their platforms have made it possible to perform cloud-based processing and machine learning classification efficiently.

Finally, I thank my institution for encouraging participation in such technical programs that bridge theoretical learning with real-world applications.

ABSTRACT

Land Use and Land Cover (LULC) mapping is a fundamental application of remote sensing and GIS, widely used for urban planning, environmental monitoring, disaster management, and sustainable land resource management. Rapid urbanization and environmental transformation require accurate and automated mapping approaches. Traditional classification techniques are often time-consuming and less efficient when dealing with large datasets.

This project focuses on AI-driven supervised classification of Sentinel-2 multispectral satellite imagery to generate an accurate LULC map for Chennai District, Tamil Nadu. The Random Forest machine learning algorithm was implemented in Google Earth Engine to classify land cover into four major categories: Water, Vegetation, Urban/Built-up, and Barren land. NDVI was calculated to enhance vegetation discrimination and improve classification accuracy.

The methodology involved satellite data acquisition, cloud filtering, composite generation, training sample collection, classifier training, classification, and accuracy assessment using confusion matrix and Kappa coefficient. QGIS was used for visualization, styling, and preparation of a professional thematic layout.

The classification achieved high overall accuracy, demonstrating the effectiveness of AI-based supervised classification in complex urban landscapes. The project highlights the integration of remote sensing and machine learning for efficient geospatial analysis.

TABLE OF CONTENTS

S. No.	Section Title	Page No.
1	Introduction	6
2	Objective	7
3	Study Area	8
4	Data Used	9-11
5	Theoretical Background	12
6	Methodology	13–16
7	Results	17-18
8	Accuracy Assessment	19
9	Area Statistics	20-21
10	Discussion	22
11	Limitations	23
12	Future Improvements	24
13	Conclusion	25
14	References	26

INTRODUCTION

Remote sensing is the science of obtaining information about objects or areas from a distance, typically through satellite or aerial imagery. Geographic Information Systems (GIS) enable the storage, analysis, and visualization of spatial data. The integration of remote sensing and GIS has revolutionized environmental monitoring and land resource management.

Land Use and Land Cover (LULC) mapping refers to the classification of Earth's surface into distinct categories such as vegetation, water bodies, urban areas, and barren land. Accurate LULC information is essential for sustainable development planning, flood risk analysis, climate change studies, and infrastructure expansion.

Chennai, being a rapidly growing metropolitan city, experiences continuous transformation in land cover due to urban expansion, industrial development, and coastal processes. Manual classification methods are insufficient to handle large satellite datasets efficiently. Artificial Intelligence (AI) and machine learning techniques provide robust solutions for automated classification with improved accuracy.

This project applies supervised machine learning, specifically the Random Forest algorithm, to classify Sentinel-2 satellite imagery and generate a reliable LULC map of Chennai district.

OBJECTIVE

The primary objective of this project is to generate an accurate Land Use and Land Cover map of Chennai District using supervised machine learning techniques applied to Sentinel-2 satellite imagery.

The specific objectives include:

- To acquire cloud-free Sentinel-2 imagery for the study area.
- To preprocess and clip imagery to the district boundary.
- To compute NDVI for vegetation enhancement.
- To collect representative training samples for each land cover class.
- To train and apply a Random Forest classifier.
- To evaluate classification accuracy using statistical metrics.
- To calculate class-wise area statistics.
- To prepare a professional GIS map layout.

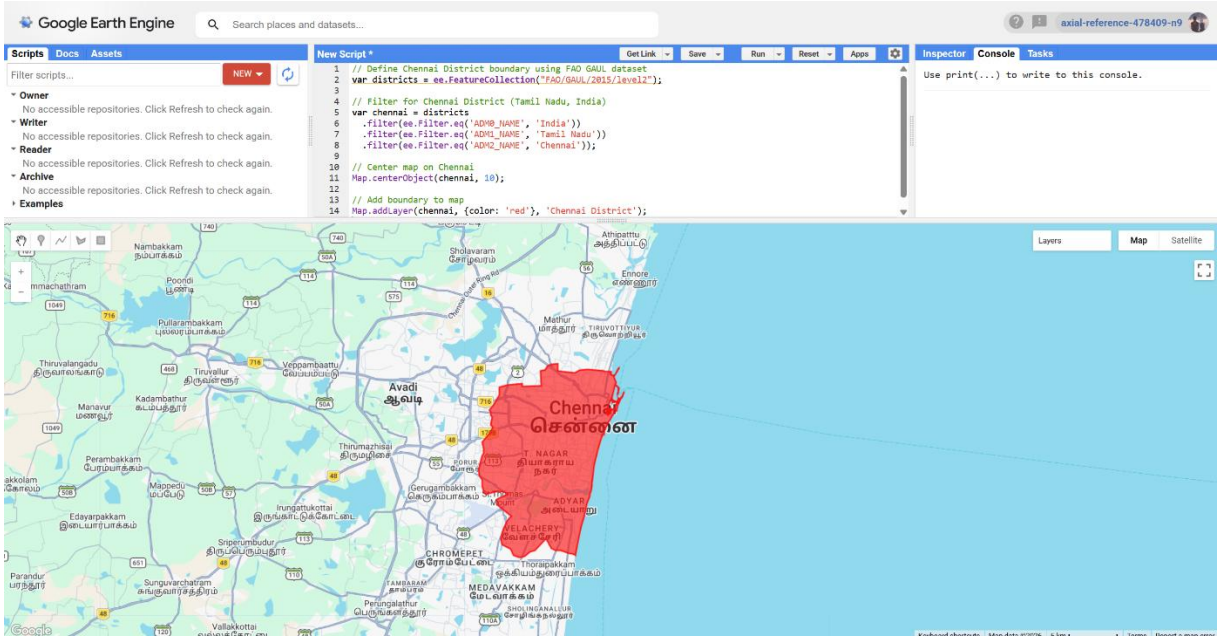
STUDY AREA

The study area selected for this project is Chennai District, located in Tamil Nadu, India. Chennai lies along the southeastern coast of India bordering the Bay of Bengal.

The district is characterized by:

- Dense urban settlements
- Major rivers such as Adyar and Cooum
- Coastal sandy stretches
- Wetlands and lakes
- Industrial and residential zones

The geographic coordinates range approximately between 12.8°N to 13.3°N latitude and 80.1°E to 80.3°E longitude.



DATA USED

Satellite Data

The primary dataset used in this study is Sentinel-2 Mul

tiSpectral Instrument (MSI) imagery. Sentinel-2 is part of the European Space Agency's Copernicus Programme and provides high-resolution optical imagery for land monitoring applications.

Dataset Used:
Sentinel-2 Surface Reflectance (Level-2A)

Platform:
Sentinel-2A / Sentinel-2B

Spatial Resolution:
10 meters (for visible and near-infrared bands)

Temporal Coverage:
January 2023 – December 2023

Cloud Filtering Criteria:
Images with Cloudy Pixel Percentage < 10%

The imagery was accessed and processed using Google Earth Engine (GEE), which enables cloud-based geospatial analysis without local data download.

Spectral Bands Used

The following Sentinel-2 bands were selected for classification due to their importance in land cover discrimination:

Band	Wavelength Region	Spatial Resolution	Purpose
Band 2	Blue	10 m	Water body detection, atmospheric correction
Band 3	Green	10 m	Vegetation analysis
Band 4	Red	10 m	Vegetation and urban detection
Band 8	Near Infrared (NIR)	10 m	Vegetation health analysis

These bands were chosen because they effectively capture spectral variations between water, vegetation, built-up areas, and barren land.

Derived Spectral Index

To enhance vegetation detection, the Normalized Difference Vegetation Index (NDVI) was calculated using:

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

Where:

- NIR = Band 8
- Red = Band 4

NDVI values range from −1 to +1. Higher values indicate healthy vegetation, while lower values represent water bodies or built-up areas. The inclusion of NDVI significantly improved vegetation classification accuracy.

Administrative Boundary Data

The boundary of Chennai District was obtained from:

FAO GAUL (Global Administrative Unit Layers) Dataset

The boundary was used to clip satellite imagery to the Area of Interest (AOI).

Data Source

The datasets used in this project were accessed through the following platforms:

- Google Earth Engine (GEE) cloud platform
- Copernicus Open Access Hub
- FAO GAUL Administrative Boundaries

All data used in this study are open-source and freely available for academic purposes.

THEORETICAL BACKGROUND

Remote Sensing Principles

Remote sensing operates based on electromagnetic radiation. Different surfaces reflect and absorb energy differently, forming unique spectral signatures. Vegetation reflects strongly in the near-infrared region, water absorbs most radiation, and urban areas show distinct reflectance patterns.

Supervised Classification

Supervised classification involves training a model using □abelled samples. The classifier learns spectral characteristics from training data and assigns each pixel to the most probable class.

Random Forest Algorithm

Random Forest is an ensemble learning algorithm that builds multiple decision trees during training. Each tree makes a prediction, and the final classification is determined through majority voting. It reduces overfitting and improves robustness.

Advantages include:

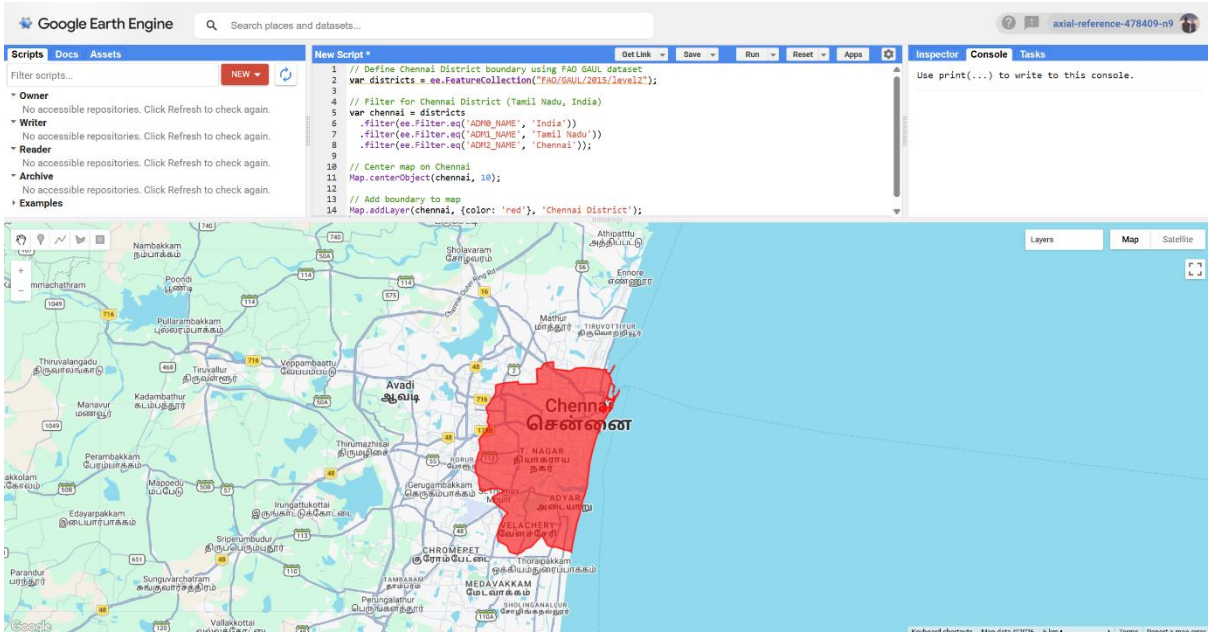
- High accuracy
- Handles large datasets
- Resistant to noise
- Suitable for multispectral imagery

METHODOLOGY

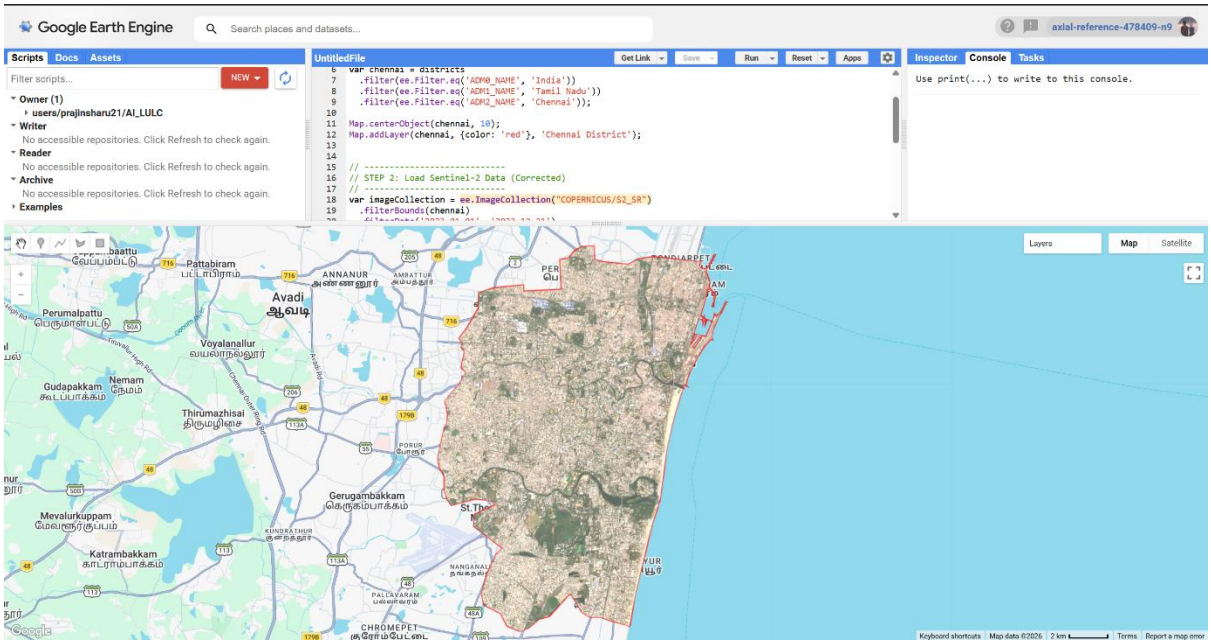
The workflow was divided into Google Earth Engine and QGIS processes.

Google Earth Engine Workflow

1. Defined Chennai district as Area of Interest.

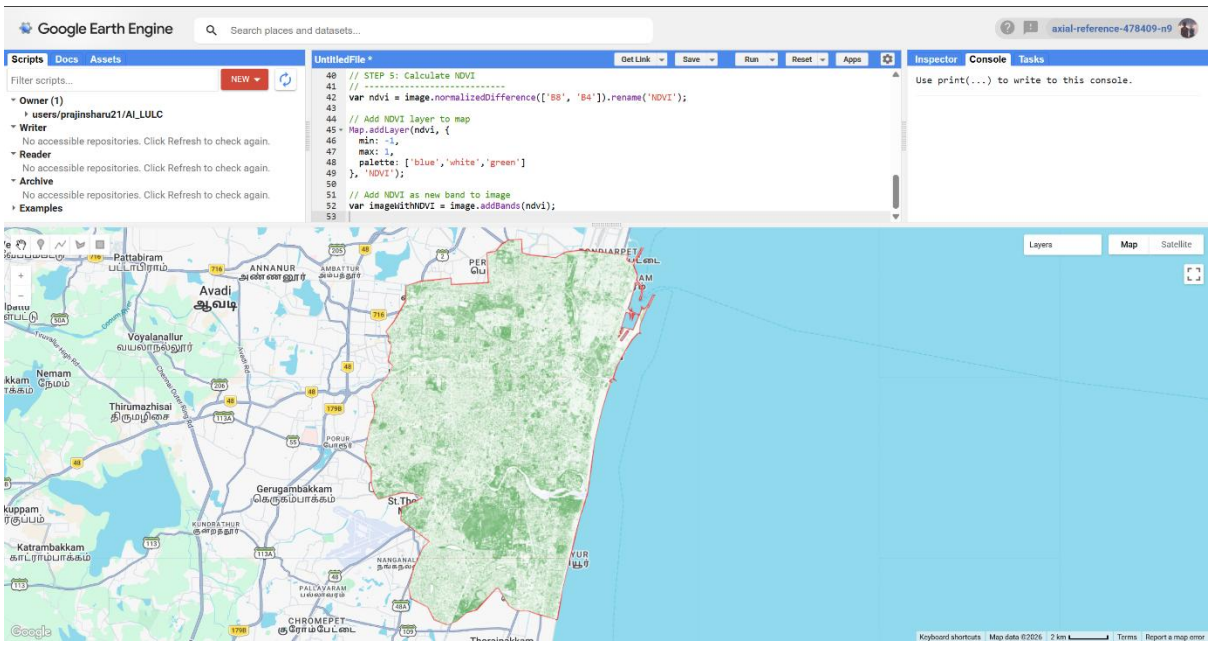


2. Filtered Sentinel-2 imagery for 2023 with cloud cover less than 10%.

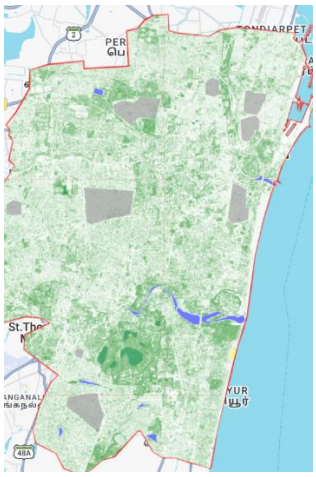


3. Created median composite.

4. Calculated NDVI.

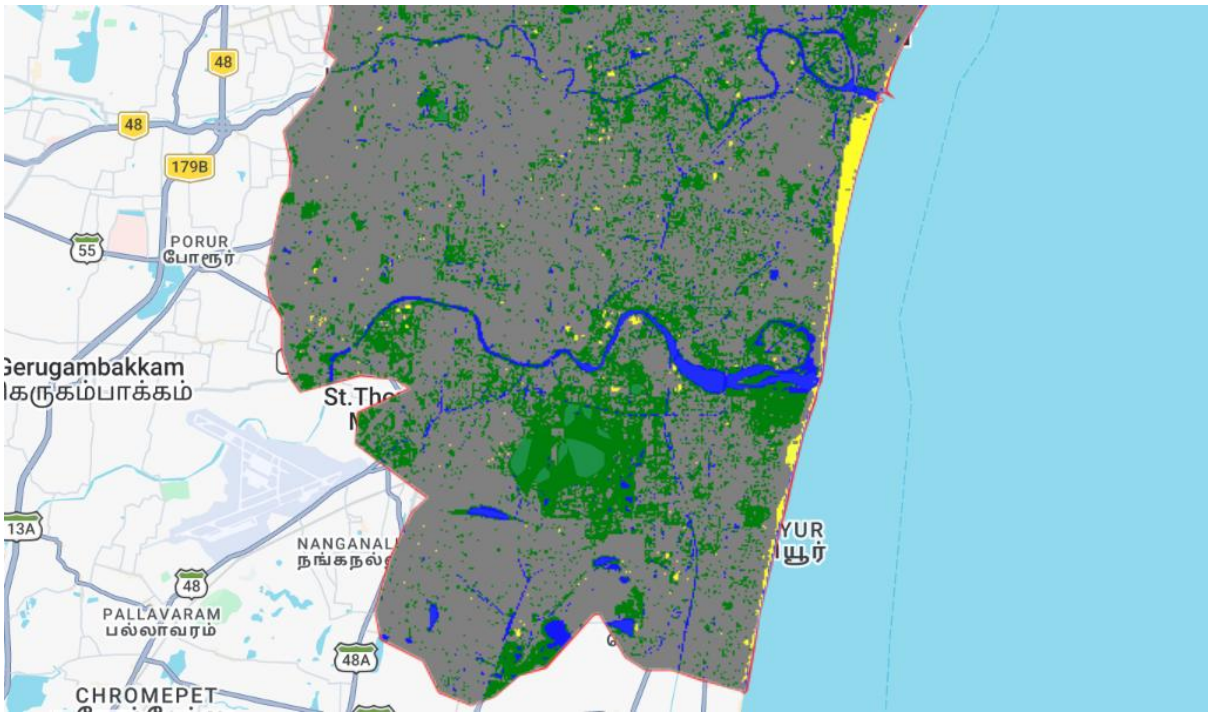


5. Collected training polygons for four classes.



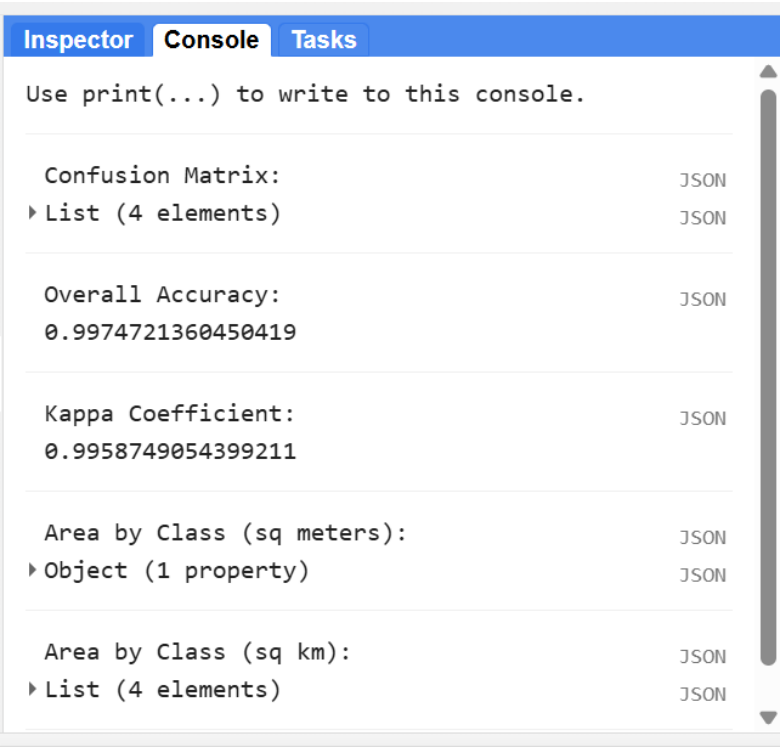
6. Trained Random Forest classifier with 50 trees.

7. Generated classified LULC raster.



8. Performed accuracy assessment.

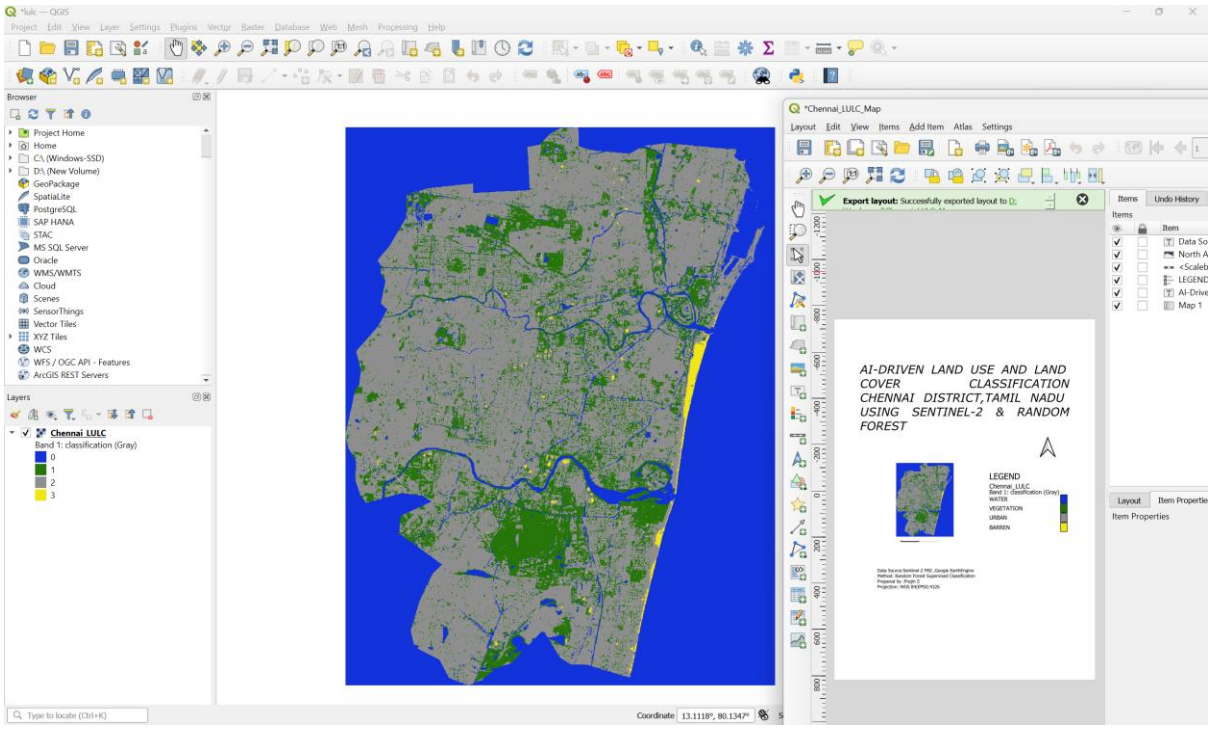
9. Calculated area statistics.



QGIS Workflow

1. Imported classified raster.
2. Applied categorized symbology.

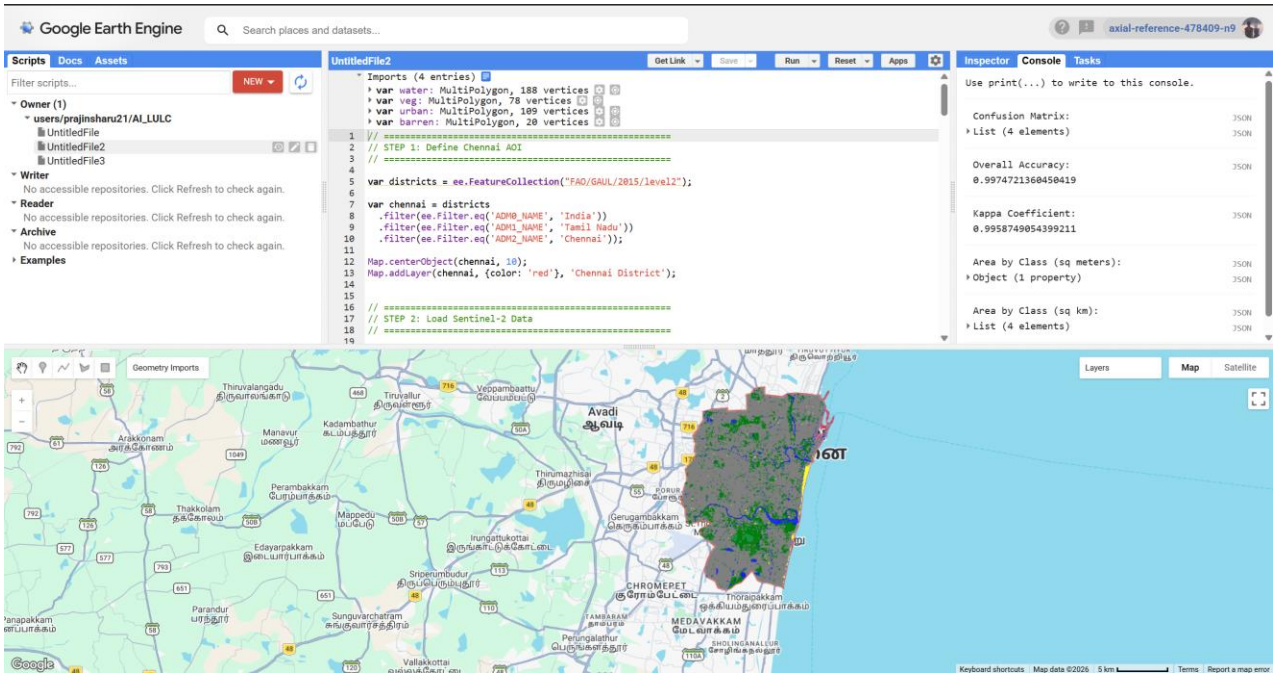
3. Assigned class colors:
    - Blue – Water
    - Green – Vegetation
    - Grey – Urban
    - Yellow – Barren
4. Created layout including title, legend, scale bar, north arrow.
5. Exported final map as PDF.



RESULTS

The final LULC classification map successfully categorized Chennai district into four major land cover classes.

Urban areas dominate central and northern Chennai. Vegetation is concentrated in southern parts and institutional campuses. Water bodies include rivers, lakes, and coastal zones. Barren land appears near sandy coastal strips.



AI-DRIVEN LAND USE AND LAND COVER CLASSIFICATION CHENNAI DISTRICT,TAMIL NADU USING SENTINEL-2 & RANDOM FOREST

LEGEND

Chennai_LULC
Band 1: classification (Gray)

WATER

VEGETATION

URBAN

BARREN

Data Source:Sentinel-2 MSI ,Google EarthEngine
Method: Random Forest Supervised Classification
Prepared by :Prajin S
Projection: WGS 84(EPSG:4326

ACCURACY ASSESSMENT

Confusion matrix analysis was performed.

Overall Accuracy: 99.74%
Kappa Coefficient: 99.58%

The high accuracy indicates strong agreement between predicted and reference classes.

AREA STATISTICS

Class-wise area distribution was calculated in square kilometers.

Class ID	Land Cover Type	Area (sq. km)
0	Water	10.131
1	Vegetation	42.514
2	Urban / Built-up	141.400
3	Barren Land	1.837


```
Area by Class (sq km):                    JSON
▾List (4 elements)                        JSON
  ▾0: Object (2 properties)
      area_sq_km: 10.131722057540111
      class: 0
  ▾1: Object (2 properties)
      area_sq_km: 42.51442525350698
      class: 1
  ▾2: Object (2 properties)
      area_sq_km: 141.40084168290275
      class: 2
  ▾3: Object (2 properties)
      area_sq_km: 1.837706116945304
      class: 3
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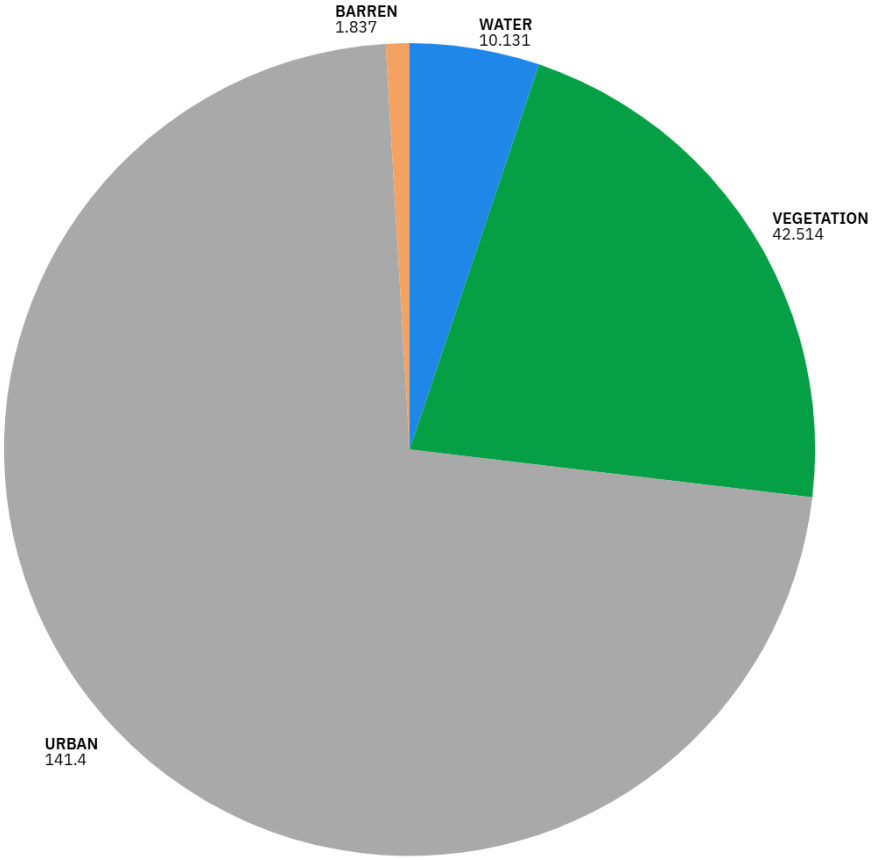

The urban class occupies the largest portion, followed by vegetation, water, and barren land.

DISCUSSION

The supervised classification results demonstrate effective separation of the major land cover types within Chennai district. The Random Forest algorithm successfully identified urban, vegetation, water, and barren classes with high overall accuracy. The dominance of the urban class aligns with the real-world land use pattern of Chennai as a metropolitan city. Vegetation areas were clearly distinguished using NDVI, which enhanced spectral differences between green cover and built-up surfaces.

Water bodies such as rivers, lakes, and the coastal region were accurately classified due to their distinct spectral reflectance characteristics. However, minor misclassification may occur in areas with mixed pixels, especially in dense urban zones where vegetation and built-up structures coexist within a single pixel. Overall, the results confirm that AI-based supervised classification provides reliable and efficient land cover mapping for complex urban environments.

LIMITATIONS

Although the classification achieved high accuracy, certain limitations must be acknowledged. The study was based on imagery from a specific time period in 2023, and seasonal variations were not considered. Vegetation reflectance changes across seasons may influence classification results. Incorporating multi-season data could provide more stable outputs.

Additionally, the number and distribution of training samples impact classification performance. Limited or uneven training data may introduce bias in certain classes. Mixed pixel effects in 10-meter resolution imagery can also cause minor confusion between urban and vegetation areas. These factors should be considered when interpreting the results.

FUTURE IMPROVEMENTS

Future improvements can focus on incorporating multi-season satellite imagery to capture temporal variations in land cover. Adding additional spectral indices such as NDBI for built-up detection or NDWI for water identification could further enhance classification accuracy.

Advanced machine learning techniques such as Support Vector Machines or deep learning-based approaches could also be explored for improved performance. Using higher spatial resolution imagery would reduce mixed pixel problems and provide more detailed urban mapping. These enhancements would strengthen the reliability and applicability of the classification model.

CONCLUSION

This project successfully implemented an AI-driven supervised classification approach to generate a Land Use and Land Cover map of Chennai district using Sentinel-2 satellite imagery. The Random Forest algorithm effectively classified the region into water, vegetation, urban, and barren categories with high accuracy.

The integration of NDVI and machine learning significantly improved class separability and mapping efficiency. The study demonstrates the practical application of remote sensing and AI techniques in urban land monitoring. The methodology can be extended for future environmental analysis, urban planning, and sustainable land management studies.

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